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SE Lab-7

FUZZ TESTING FOR INPUT VALIDATION BUGS

1)Original application execution

```
(venv) aditi@Aditis-MacBook-Air-2 Lab 7 % python loan_eligibility.py
==== Running parse_income ====
Enter monthly income (e.g. 50,000): 50,000
Income: 50000

==== Running monthly_emi ====
Enter principal,rate,years (e.g. 500000,10,5): 500000,10,5
Monthly EMI: 10623.52

==== Running pan_category ====
Enter a PAN number (e.g. ABCPE1234F): ABCPE1234F
PAN Category: Individual
(venv) aditi@Aditis-MacBook-Air-2 Lab 7 %
```

2) Failing parse_income test

```
(venv) aditi@Aditis-MacBook-Air-2 Lab 7 % python -m pytest test_loan_eligibility.py -v
============================= test session starts ==============================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0 -- /Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/venv/bin/python3
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7
plugins: hypothesis-6.141.1
collected 3 items

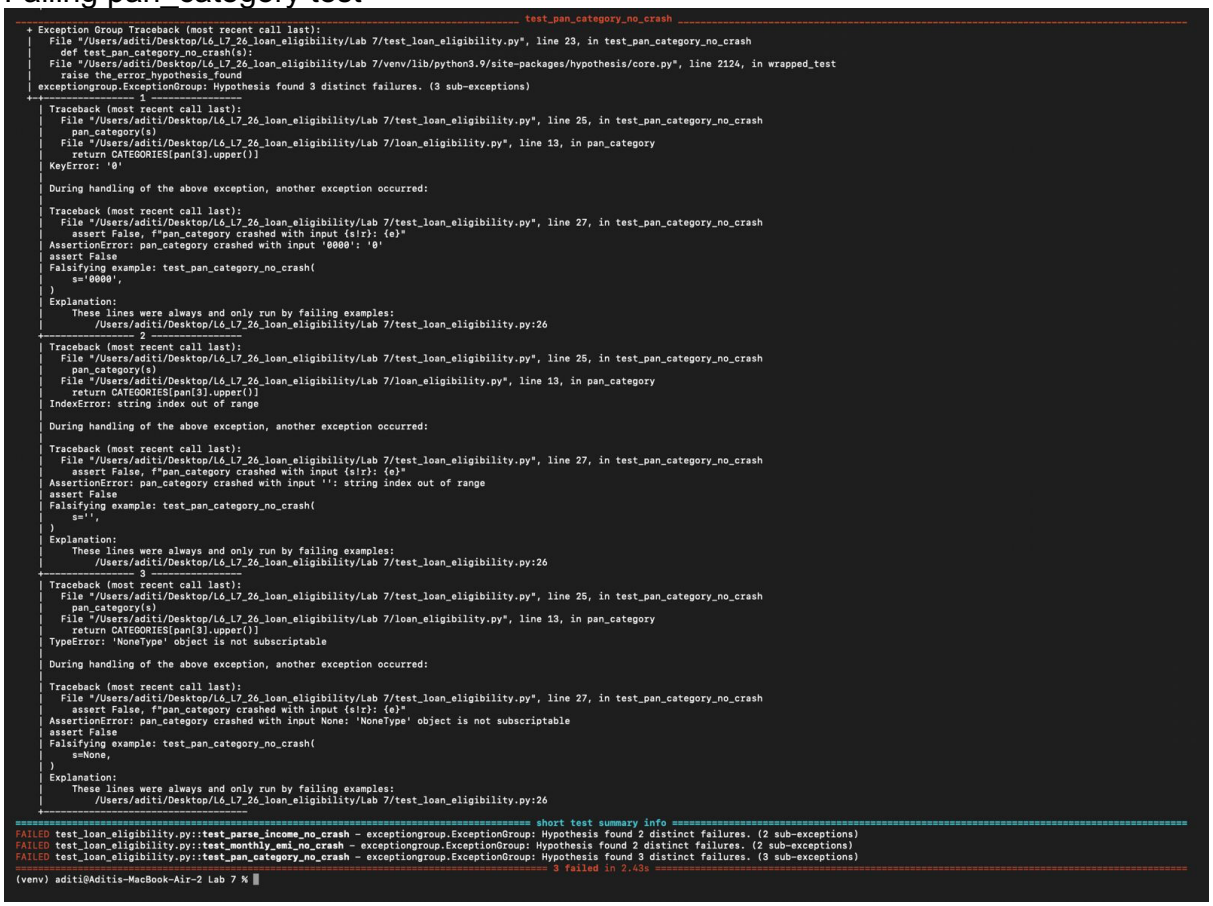
test_loan_eligibility.py::test_parse_income_no_crash FAILED              [ 33%]
test_loan_eligibility.py::test_monthly_emi_no_crash FAILED               [ 66%]
test_loan_eligibility.py::test_pan_category_no_crash FAILED              [100%]

=================================== FAILURES ===================================
_____________________________ test_parse_income_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 7, in test_parse_income_no_crash
  |     def test_parse_income_no_crash(s):
  |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/venv/lib/python3.9/site-packages/hypothesis/core.py", line 2124, in wrapped_test
  |     raise the_error_hypothesis_found
  | exceptiongroup.ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 9, in test_parse_income_no_crash
    |     parse_income(s)
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/loan_eligibility.py", line 4, in parse_income
    |     return int(income.replace(',', ''))
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 11, in test_parse_income_no_crash
    |     assert False, f"parse_income crashed with input {s!r}: {e}"
    | AssertionError: parse_income crashed with input '': invalid literal for int() with base 10: ''
    | assert False
    | Falsifying example: test_parse_income_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         /Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py:10
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 9, in test_parse_income_no_crash
    |     parse_income(s)
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/loan_eligibility.py", line 4, in parse_income
    |     return int(income.replace(',', ''))
    | AttributeError: 'NoneType' object has no attribute 'replace'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 11, in test_parse_income_no_crash
    |     assert False, f"parse_income crashed with input {s!r}: {e}"
    | AssertionError: parse_income crashed with input None: 'NoneType' object has no attribute 'replace'
    | assert False
    | Falsifying example: test_parse_income_no_crash(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         /Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py:10
    +------------------------------------
```

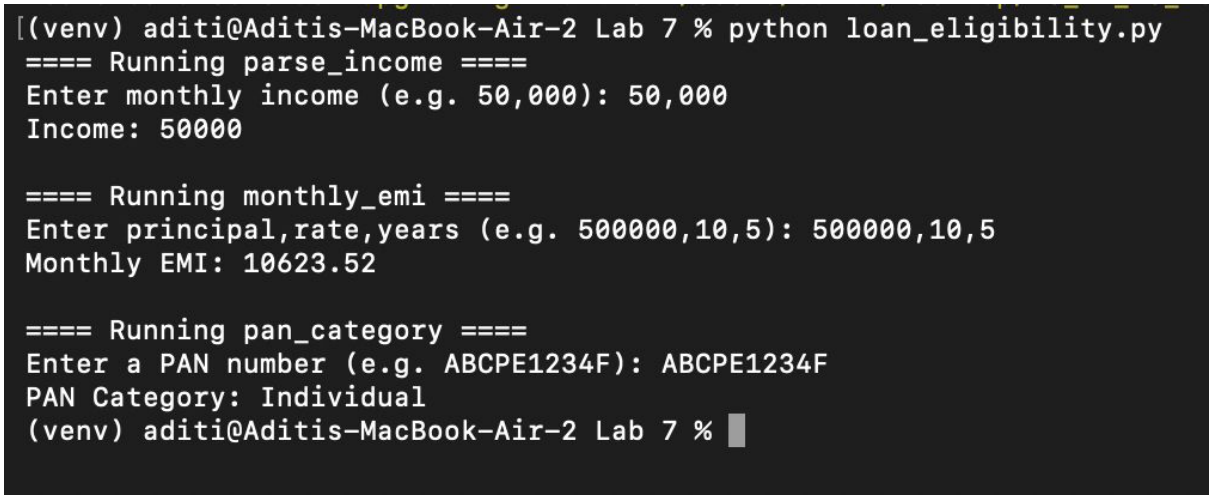
Failing monthly_emi test

```
_____________________________ test_monthly_emi_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 15, in test_monthly_emi_no_crash
  |     def test_monthly_emi_no_crash(s):
  |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/venv/lib/python3.9/site-packages/hypothesis/core.py", line 2124, in wrapped_test
  |     raise the_error_hypothesis_found
  | exceptiongroup.ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 17, in test_monthly_emi_no_crash
    |     monthly_emi(s)
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/loan_eligibility.py", line 7, in monthly_emi
    |     principal, rate, years = details.split(',')
    | ValueError: not enough values to unpack (expected 3, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 19, in test_monthly_emi_no_crash
    |     assert False, f"monthly_emi crashed with input {s!r}: {e}"
    | AssertionError: monthly_emi crashed with input '': not enough values to unpack (expected 3, got 1)
    | assert False
    | Falsifying example: test_monthly_emi_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 17, in test_monthly_emi_no_crash
    |     monthly_emi(s)
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/loan_eligibility.py", line 7, in monthly_emi
    |     principal, rate, years = details.split(',')
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aditi/Desktop/L6_L7_26_loan_eligibility/Lab 7/test_loan_eligibility.py", line 19, in test_monthly_emi_no_crash
    |     assert False, f"monthly_emi crashed with input {s!r}: {e}"
    | AssertionError: monthly_emi crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Falsifying example: test_monthly_emi_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________ test_pan_category_no_crash _____________________________
```

Failing pan_category test



3) Fixed application execution



4) Passing Hypothesis tests

